

Cheatsheet

Probabilities

Global conventions

- Random variables are capital letters (e.g., X, S, R); values are lowercase (e.g., x, s, r).
- $\log \equiv \log_2$. Units: bits. Natural logs use $\ln$ and give nats.
- A probability density function (PDF) $p_X(x)$ satisfies $p_X(x) \geq 0$ and $\int_x p_X(x) = 1$.

Sample spaces, PDFs, and expectations

- Finite set (sample space) Ω ; cardinality $|\Omega|$.
- Random variable $X : \Omega \rightarrow \mathcal{X}$ and its PDF $p(x) = \Pr(X = x)$, where $\int_x p(x) = 1$ always holds and $\Pr(X \in A) = \int_{x \in A} p(x)$
- Expectation: $\mathbb{E}[f(X)] = \int_{x \in \mathcal{X}} f(x) p(x)$.
- Variance: $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

1 Notions of information theory

Joint, marginal, conditional; Bayes; independence

- Joint PDF $p(x, y) = \Pr(X = x, Y = y)$.
- Marginals: $p(x) = \int_y p(x, y)$, $p(y) = \int_x p(x, y)$.
- Conditional: $p(x | y) = \frac{p(x, y)}{p(y)}$ (when $p(y) > 0$).
- Bayes theorem:

$$p(x, y) = p(x | y)p(y) = p(y | x)p(x) \tag{1}$$

$$p(x | y) = \frac{p(y | x)p(x)}{p(y)} \tag{2}$$

- Independence: $X \perp Y \iff p(x, y) = p(x)p(y) \quad \forall x, y \iff p(x \mid y) = p(x) \text{ for all } y \text{ with } p(y) > 0.$

Math

Logarithms and “bits” intuition

- Log rules: $\log(ab) = \log a + \log b$, $\log(a/b) = \log a - \log b$, $\log(a^k) = k \log a$.
- Change of base: $\log_b x = \frac{\log x}{\log b}$.
- Bits intuition: $\log_2 N$ is the number of yes/no questions needed to identify one of N equiprobable outcomes.

Optimization

Calculus

Find critical points by solving $f'(x) = 0$ (and where $f'(x)$ is undefined). Classify at x^* using $f''(x^*)$: > 0 min, < 0 max, $= 0$ inconclusive. On a closed interval, also check endpoints and compare function values. For a teaching example see Problem 1A.

Constrained extrema

If the optimization is subject to equality constraints, use the Lagrange multiplier method. For a teaching example see Problem 1B and for .

Functions

Min/max $f(x)$ s.t. $g_i(x) = 0$. Define

$$\mathcal{L}(x, \lambda) = f(x) + \sum_i \lambda_i g_i(x).$$

Solve $\nabla_x \mathcal{L} = 0$ together with $g_i(x) = 0$.

Functionals

Extremize $\mathcal{F}[f]$ subject to $\mathcal{G}[f] = 0$ (with $\delta \mathcal{G}[f^*] \neq 0$). Introduce the augmented functional

$$\mathcal{L}[f, \lambda] = \mathcal{F}[f] + \lambda \mathcal{G}[f].$$

At an extremum (f^*, λ^*) one has

$$\delta_f \mathcal{L}[f^*, \lambda^*] = 0 \iff \frac{\delta \mathcal{F}}{\delta f}(f^*) + \lambda^* \frac{\delta \mathcal{G}}{\delta f}(f^*) = 0,$$

i.e. the Euler–Lagrange equation for the integrand of $\mathcal{L}$ (plus boundary/transversality conditions). Choose λ^* so that $\mathcal{G}[f^*] = 0$.